

Wireless LAN I

IEEE 802.11 Basics

Overview

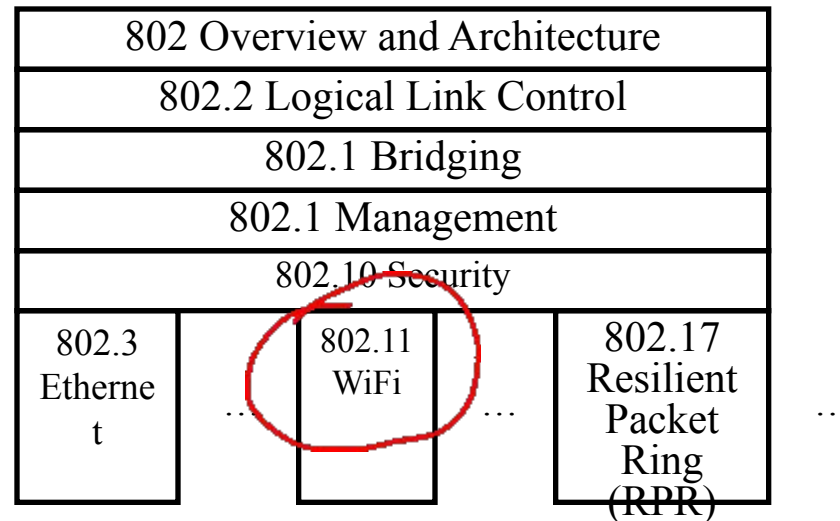
1. IEEE 802.11 vs. WiFi
2. IEEE Standards Numbering System
3. Key features of 802.11
4. 802.11 Bands and Channels
5. Hidden Node Problem and 4-Way Handshake RTS/CTS)
6. 802.11 MAC (inter-frame space, PCF, DCF)
7. 802.11 Architecture and Addressing
8. 802.11 Frame Format
9. 802.11 Power Management

IEEE 802.11 vs. WiFi

- ❑ IEEE 802.11 is a standard
- ❑ WiFi = “Wireless Fidelity” is a trademark
- ❑ Fidelity = **Compatibility** between wireless equipment from different manufacturers
- ❑ WiFi Alliance is a non-profit organization that does the compatibility testing (WiFi.org)
- ❑ 802.11 has many options and it is possible for two equipment based on 802.11 to be *incompatible*.
- ❑ All equipment with “WiFi” logo have selected options such that they will **interoperate**.

IEEE Standards Numbering System

- ❑ IEEE 802.* and IEEE 802.1* standards (e.g., IEEE 802.1Q-2011) apply to all IEEE 802 technologies:
 - ❑ IEEE 802.3 Ethernet
 - ❑ IEEE 802.11 WiFi
 - ❑ IEEE 802.16 WiMAX



Lettered vs. Numerical Versions

- ❑ IEEE 802.11 uses *letters* to name the versions
 - E.g., 802.11a/b (1999), 802.11g (2003), 802.11n (2009), 802.11ac (2013), and so on
- ❑ WiFi Alliance proposes numbers to simplify
 - E.g., WiFi 4 (802.11n), WiFi 5 (802.11ac), WiFi 6 (802.11ax) ...

IEEE 802.11 Features

- ❑ Data rate (a.k.a. *speed*)
 - ❑ Original IEEE 802.11-1997 was at 1 and 2 Mbps.
 - ❑ Newer versions at 11 Mbps, 54 Mbps, 108 Mbps, 1.2 Gbps, ...
- ❑ Spectrum licensing
 - ❑ All versions use *license-exempt* spectrum
- ❑ PHYs:
 - ❑ Spread spectrum (in old versions)
 - ❑ OFDM (in new versions)
- ❑ Supports multiple priorities (time-critical and data traffic)
- ❑ Power management allows a node to 'doze off'
 - ❑ Longer battery life

IEEE 802.11 Physical Layers

- ❑ Issued in several stages
- ❑ First version in 1997: Legacy IEEE 802.11 (no longer used)
 - ❑ 3 physical layer specifications (2 in 2.4-GHz, 1 in infrared)
 - ❑ All operating at 1 and 2 Mbps
- ❑ Amendments in 1999:
 - ❑ IEEE 802.11a-1999: 5-GHz band, 54 Mbps/20 MHz, **OFDM**
 - ❑ IEEE 802.11b-1999: 2.4 GHz band, 11 Mbps/22 MHz (spread spectrum)
- ❑ Amendment in 2003:
 - ❑ IEEE 802.11g-2003 : 2.4 GHz band, 54 Mbps/20 MHz, **OFDM**

ISM Bands

- Industrial, Scientific, and Medical bands. License exempt

From	To	Bandwidth	Availability
6.765 MHz	6.795 MHz	30 kHz	
13.553 MHz	13.567 MHz	14 kHz	Worldwide
26.957 MHz	27.283 MHz	326 kHz	Worldwide
40.660 MHz	40.700 MHz	40 kHz	Worldwide
433.050 MHz	434.790 MHz	1.74 MHz	Europe, Africa, Middle east, Former Soviet Union
902.000 MHz	928.000 MHz	26 MHz	America, Greenland
2.400 GHz	2.500 GHz	100 MHz	Worldwide
5.725 GHz	5.875 GHz	150 MHz	Worldwide
24.000 GHz	24.250 GHz	250 MHz	Worldwide
61.000 GHz	61.500 GHz	500 MHz	
122.000 GHz	123.000 GHz	1 GHz	
244 GHz	246 GHz	2 GHz	

WLAN/WiFi Bands

WLAN/WiFi Standard	Frequency Band
802.11b/g/n	2.4 GHz
802.11a/n/ac/ax	5 GHz
802.11be	6 GHz (not confirmed yet)
802.11p (car-to-car)	5.9 GHz (licensed band)
802.11ah (IoT)	900 MHz
802.11af (Rural)	700 MHz (unused TV channels)
802.11ad/ay (Multi Gbps wireless applications: e.g., cable replacement, VR, ...)	60 GHz

WiFi Channels

- ❑ The entire *band* is divided into several individual *channels*
- ❑ An AP operates over a **single channel** at any given time
- ❑ Different nearby APs can operate over different channels of the same band
 - ❑ Avoid congestion and interference
- ❑ Each channel is usually 20 or 22 MHz wide
- ❑ With newer WiFi versions, it is possible to combine two or more channels to get a wider channel
 - ❑ More bandwidth for higher data rates

2.4 GHz WiFi Channel Frequencies

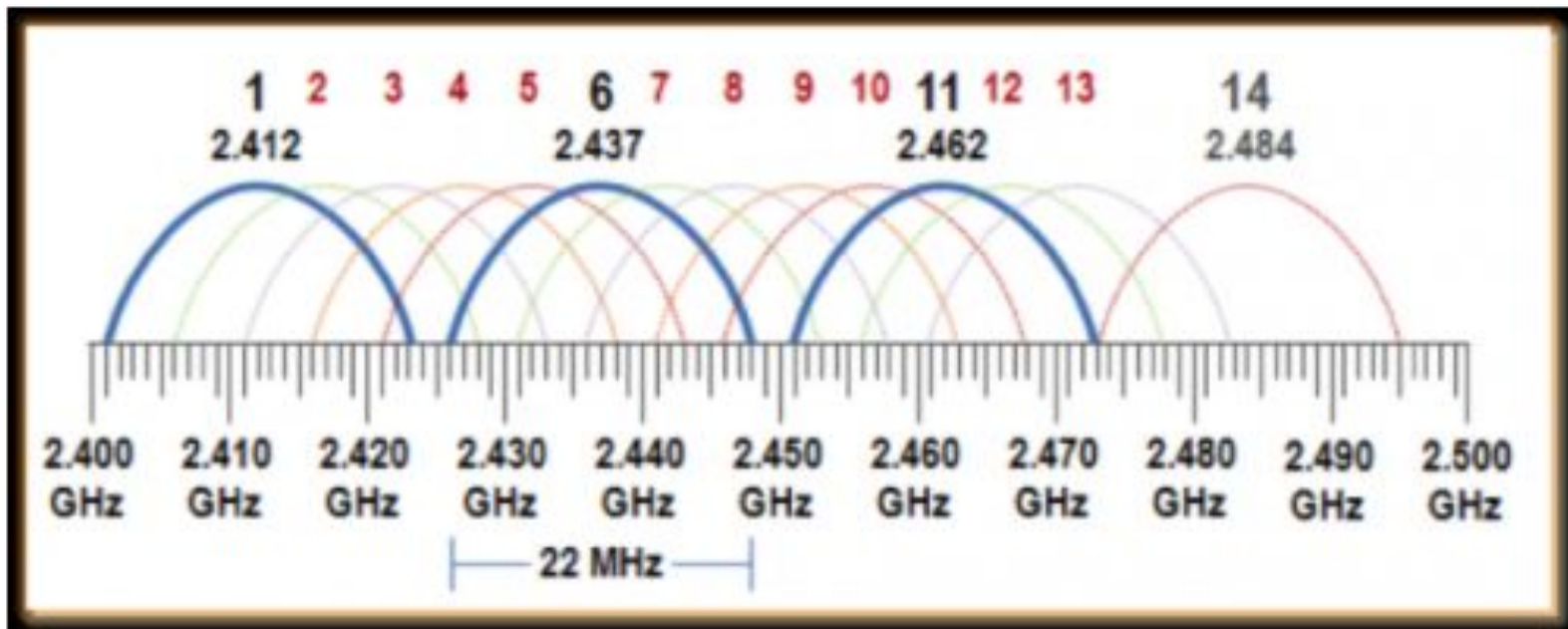
- ❑ A total of 14 channels (not all channels available in all countries)
- ❑ Centre frequencies are 5 MHz apart (except channel 14)
- ❑ Each channel is 22 MHz wide

CHANNEL NUMBER	LOWER FREQUENCY MHZ	CENTER FREQUENCY MHZ	UPPER FREQUENCY MHZ
1	2401	2412	2423
2	2406	2417	2428
3	2411	2422	2433
4	2416	2427	2438
5	2421	2432	2443
6	2426	2437	2448
7	2431	2442	2453
8	2436	2447	2458
9	2441	2452	2463
10	2446	2457	2468
11	2451	2462	2473
12	2456	2467	2478
13	2461	2472	2483
14	2473	2484	2495

From <http://www.radio-electronics.com/info/wireless/wi-fi/80211-channels-number-frequencies-bandwidth.php>

2.4 GHz Channel Overlaps

- ❑ Most channels in 2.4 GHz band overlap
- ❑ Maximum of **three non-overlapping channels** are possible
- ❑ 1-6-11 are most widely used non-overlapping channels (6 is usually default)
 - ❑ E.g., if three are three nearby APs in an enterprise, they are usually set to 1-6-11



From <http://boundless.aerohive.com/experts/WLAN-Channels-Explained.html>

Channels in 5 GHz Band

- ❑ 20 MHz channels (c.f. 22 MHz in 2.4 GHz band)
- ❑ Non-overlapping (c.f., mostly overlapping in 2.4 GHz)
- ❑ Two types of channels
 - ❑ Always available
 - ❑ Channels used by **radar** (requires DFS)
- ❑ Dynamic Frequency Selection (DFS): WiFi AP monitors radar channels and vacate them (switch to another channel) if radar is detected
 - ❑ May cause connection drop for clients

5GHz Channel Structure

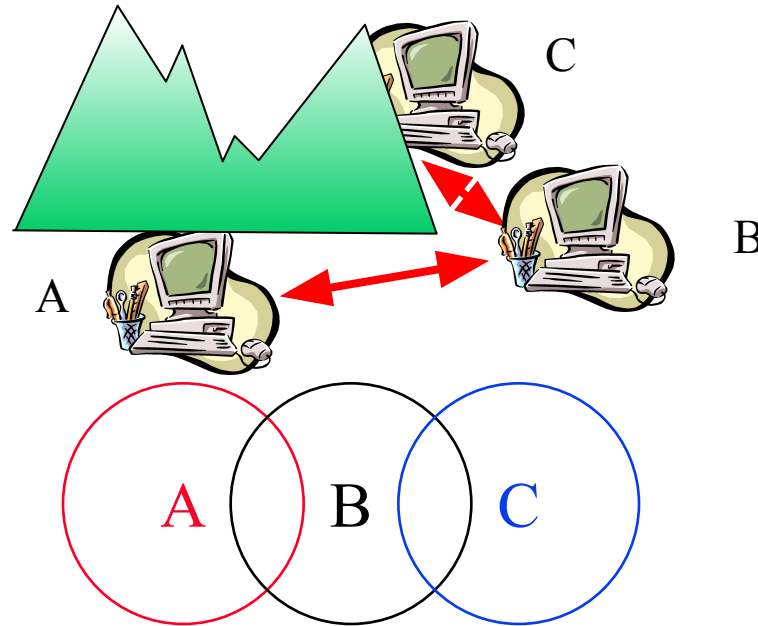
5 GHz Channel Allocations

Frequency (GHz)	5.150				5.250				5.470				5.600				5.640				5.725				5.850			
													TDWR															
802.11 Allocations	UNII-1				UNII-2a				UNII-2c (Extended)												UNII-3							
Center Frequency	5180	5200	5220	5240	5260	5280	5300	5320	5500	5520	5540	5560	5580	5600	5620	5640	5660	5680	5700	5720	5745	5765	5785	5805	5825			
20 MHz	36	40	44	48	52	56	60	64	100	104	108	112	116	120	124	128	132	136	140	144	149	153	157	161	165			
40 MHz	38		46		54		62		102		110		118		126		134		142		151		159					
80 MHz	42				58				106				122		138				155									
160 MHz	50								114																			
FCC	1,000 mW Tx Power Indoor & Outdoor No DFS needed				250 mw w/6dBi Indoor & Outdoor DFS Required				250mw w/6dBi Indoor & Outdoor DFS Required 144 Now Allowed				120, 124, 128 Devices Now Allowed								1,000 mW EIRP Indoor & Outdoor No DFS needed 165 was ISM, now UNII-3							
DFS Channels									DFS Channels																			

Source: *Wireless LAN Professionals*

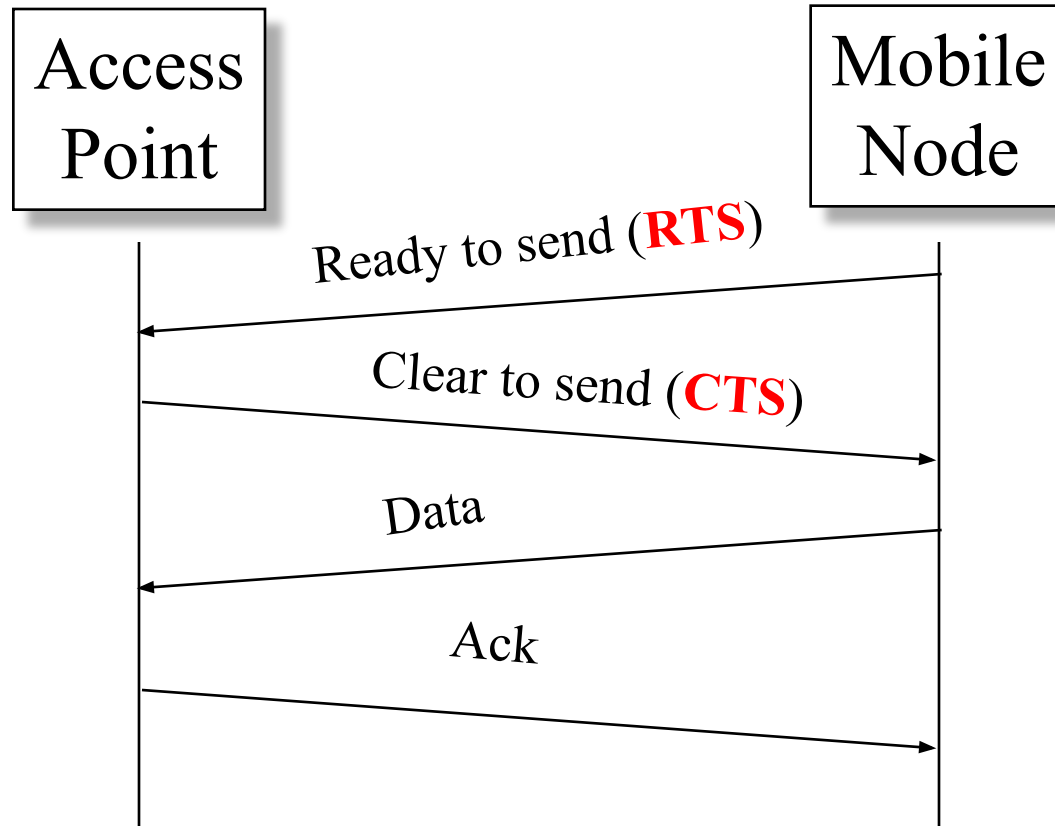
Source: <https://www.ekahau.com/blog/channel-planning-best-practices-for-better-wi-fi/> (accessed 15 June 2020): this structure is probably for the US; radar channels may vary with countries

Hidden Node Problem



- ❑ A can hear B, B can hear C, but C cannot hear A (C and A are *hidden* from each other)
- ❑ C may start transmitting while A is also transmitting ❑ collision at B!
 - ❑ A and C (wireless transmitters) can't detect collision (why?).
- ❑ CSMA/CD is not possible (CD = collision detection; CD used in Ethernet)
 - ❑ in WLAN, only the receiver can help *avoid* collisions
- ❑ **4-way handshake** needed to implement CA (*collision avoidance*) in WLAN

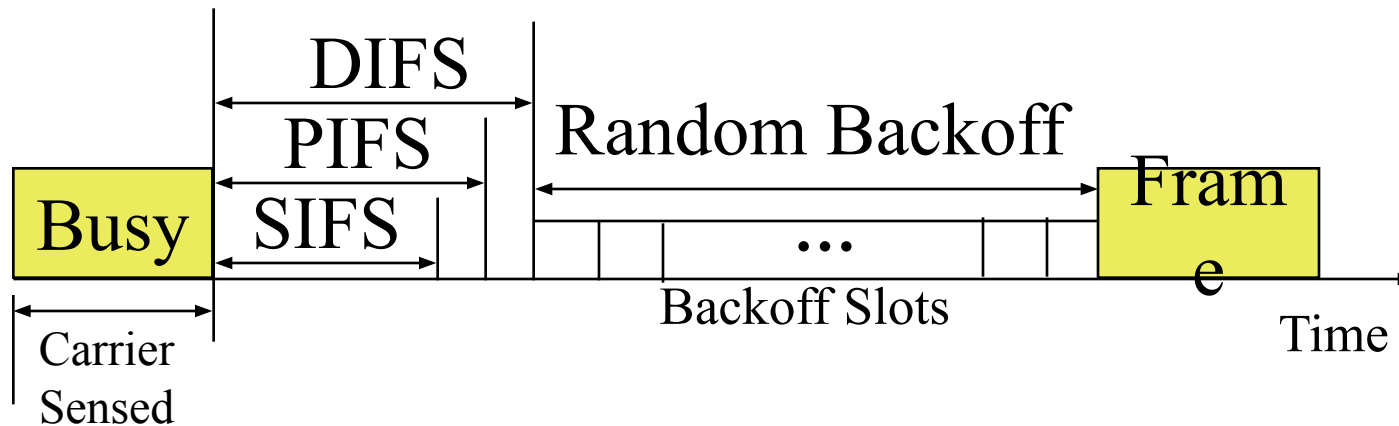
4-Way Handshake



IEEE 802.11 MAC

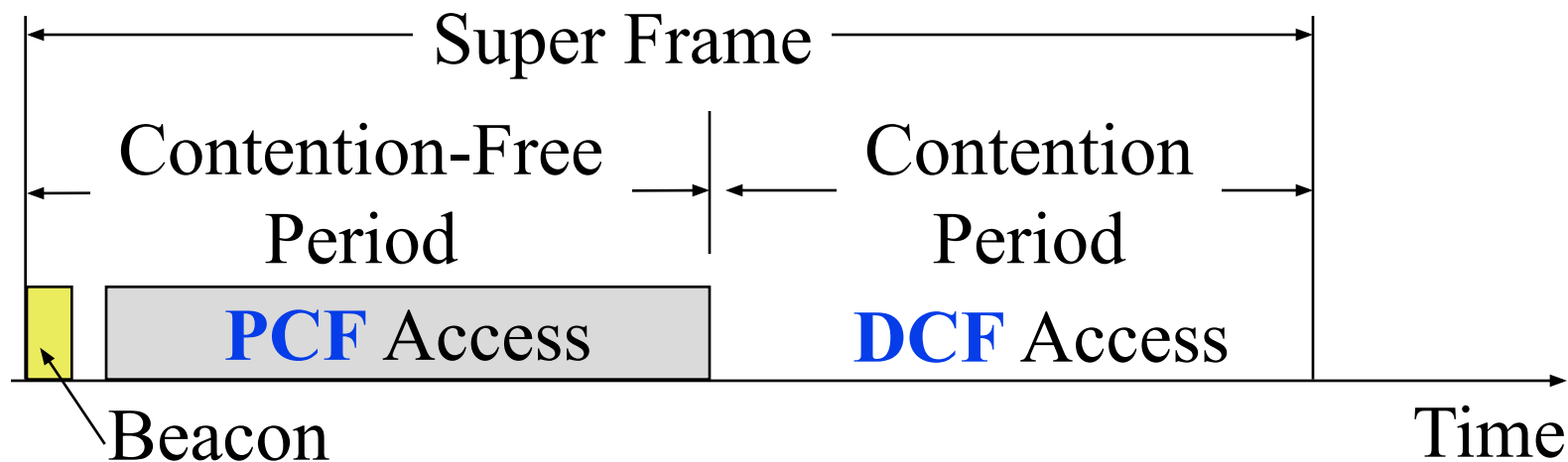
- ❑ Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)
- ❑ Listen before you talk. If the medium is busy, the transmitter backs off for a random period.
- ❑ Avoids collision by sending a short message:
Ready to send (RTS)
RTS contains dest. address and duration of message.
Tells everyone to backoff for the duration.
- ❑ Destination sends: Clear to send (CTS)
Other stations set their network allocation vector (NAV) and wait for that duration
- ❑ Cannot detect collision, hence each packet is acked.
- ❑ MAC-level retransmission if not acked.

IEEE 802.11 Priorities with Inter-frame space



- ❑ 802.11 has different priorities for **control**, **data**, and **time-critical** packets
- ❑ Achieve priorities by using different amounts of interframe space (IFS)
- ❑ Highest priority frames, e.g., Acks, use short IFS (**SIFS**)
- ❑ Medium priority time-critical frames use “Point Coordination Function IFS” (**PIFS**)
- ❑ Asynchronous data frames use “Distributed coordination function IFS” (**DIFS**)

Time Critical Services



- ❑ Timer critical services use **Point Coordination Function**
- ❑ The point coordinator allows only one station to access
- ❑ Coordinator sends a beacon frame to all stations. Then uses a polling frame to allow a particular station to have contention-free access
- ❑ Contention Free Period (CFP) varies with the load.

IEEE 802.11 DCF Backoff

- ❑ MAC works with a single FIFO Queue
 - ❑ Focuses on transmitting the packet at the head of the queue
- ❑ Three variables:
 - ❑ Contention Window (CW)
 - ❑ Backoff count (BO)
 - ❑ Network Allocation Vector (NAV)
- ❑ If a frame (RTS, CTS, Data, Ack) is heard, NAV is set to the duration in that frame. Stations sense the media after NAV expires.
- ❑ If the medium is idle for DIFS, and backoff (BO) is not already active, the station draws a random BO in $[0, CW]$ and sets the backoff timer.
 - ❑ CW is in units of *slot time* (slot time varies with 802.11 standards)
- ❑ If the medium becomes busy during backoff, the timer is paused and a new NAV is set. After NAV, back off continues.

IEEE 802.11 DCF Backoff (Cont)

- ❑ $BO = \text{random}(0, CW)$
- ❑ Initially and after each *successful* transmission:

$$CW = CW_{\min}$$

- ❑ After each *unsuccessful* attempt

$$CW = \min\{2CW + 1, CW_{\max}\}$$

Example

- Assume that we have $CW_{min}=3$ and $CW_{max}=127$ configured for a given WLAN. What would be the values of CW if there were 8 successive unsuccessful attempts after initializing the network?

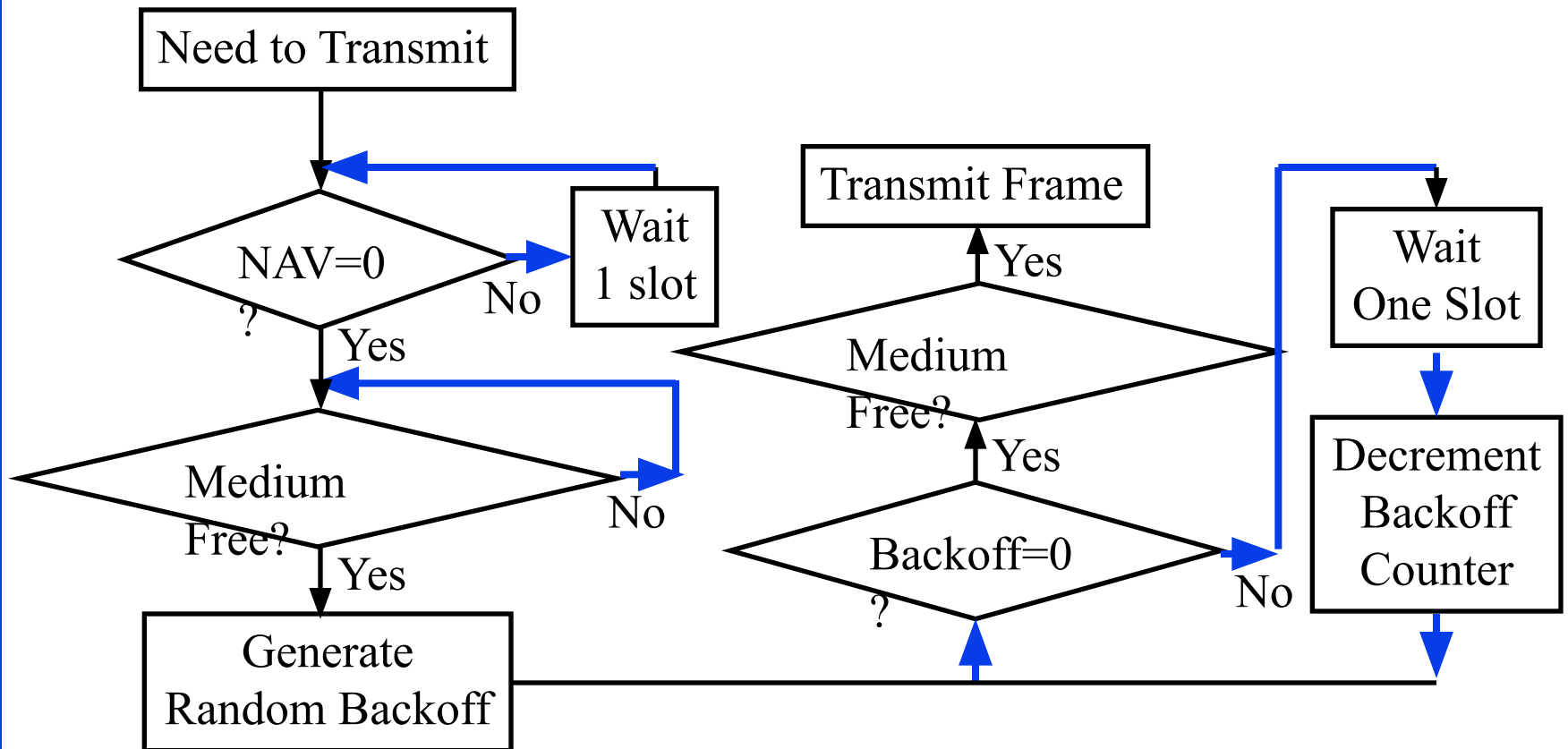
After initialization, $CW = CW_{min} = 3$

After 1st unsuccessful attempt, $CW = \min(7, 127) = 7$

After 2nd unsuccessful attempt, $CW = \min(15, 127) = 15$

Then on, 31, 63, 127, 127, 127, ...

DCF Medium Access Flow Chart



Ref: P. Roshan and J. Leary, "802.11 Wireless LAN Fundamentals," Cisco Press, 2003, ISBN:1587050773, Safari book

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Parameter Values: interframe space and contention window

WLAN	Slot-time (μ s)	SIFS (μ s)	CW _{min}	CW _{max}
11a	9	16	15	1023
11b	20	10	31	1023
11g	9 or 20	10	15 or 31	1023
11n (2.4 GHz)	9 or 20	10	15	1023
11n (5 GHz)	9	16	15	1023
11ac	9	16	15	1023

- ❑ $\text{PIFS} = \text{SIFS} + 1 \text{ slot time}$
- ❑ $\text{DIFS} = \text{SIFS} + 2 \text{ slot times} = \text{PIFS} + 1 \text{ slot time}$

Slot time: basic unit of backoff algorithm

Example

- ❑ What is the duration of PIFS and DIFS for IEEE 802.11b?

Slot time = $20 \mu\text{s}$

SIFS = $10 \mu\text{s}$

PIFS = SIFS + slot time = $10 + 20 = 30 \mu\text{s}$

DIFS = SIFS + 2 x slot time = $10 + 40 = 50 \mu\text{s}$

Virtual Carrier Sense

- ❑ Every frame has a “Duration ID” which indicates how long the medium will be busy.
 - ❑ RTS has duration of $\text{RTS} + \text{SIF} + \text{CTS} + \text{SIF} + \text{Frame} + \text{SIF} + \text{Ack}$
 - ❑ CTS has duration of $\text{CTS} + \text{SIF} + \text{Frame} + \text{SIF} + \text{Ack}$
 - ❑ Frame has a duration of $\text{Frame} + \text{SIF} + \text{ACK}$
 - ❑ ACK has a duration of ACK
 - ❑ A station has to estimate the durations of RTS/CTS/ACK
- ❑ All stations keep a “**Network Allocation Vector (NAV)**” timer in which they record the duration of each frame they hear.
- ❑ Stations do not need to sense the channel until NAV becomes zero (conserve power)

Example

- Consider an 802.11b WLAN. A station estimates the transmission times of RTS, CTS, and ACK as 10 μ s, 10 μ s, and 25 μ s, respectively. What would be the value of the Duration field in the RTS header if the station wants to send a 250 μ s long data frame ?

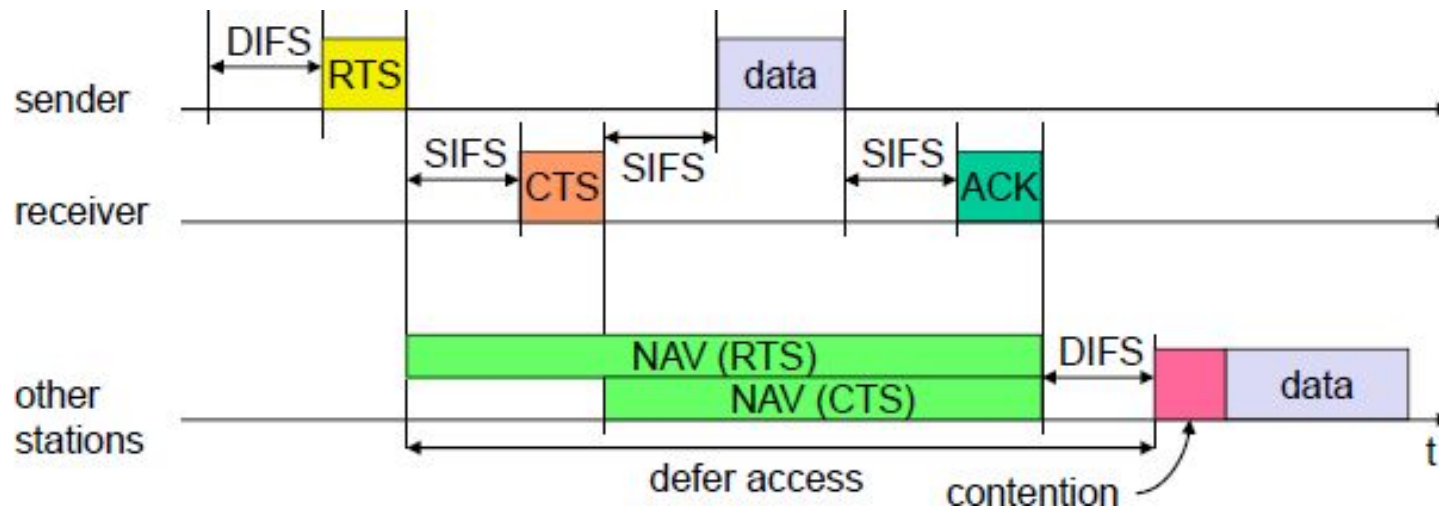
802.11b has a SIFS duration of 10 μ s.

Duration field in RTS = RTS_time + CTS_time + ACK_time + data_time + 3xSIFS

$$= 10+10+25+250+3 \times 10 = 325 \mu\text{s}$$

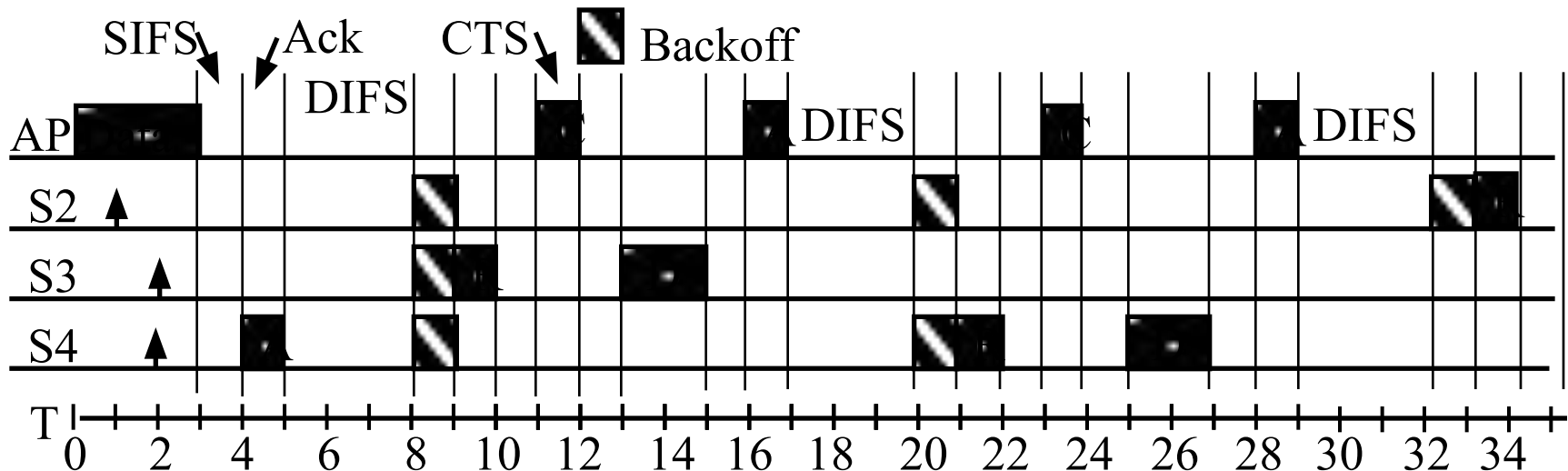
802.11 with RTS/CTS

When a node is sensing the channel, it must be free for DIFS period. SIFS is used as the wait time between the RTS, CTS, DATA and ACK frames. $SIFS < DIFS$ means that another node cannot incorrectly determine that the channel is idle during the 4-way handshake between two other nodes.



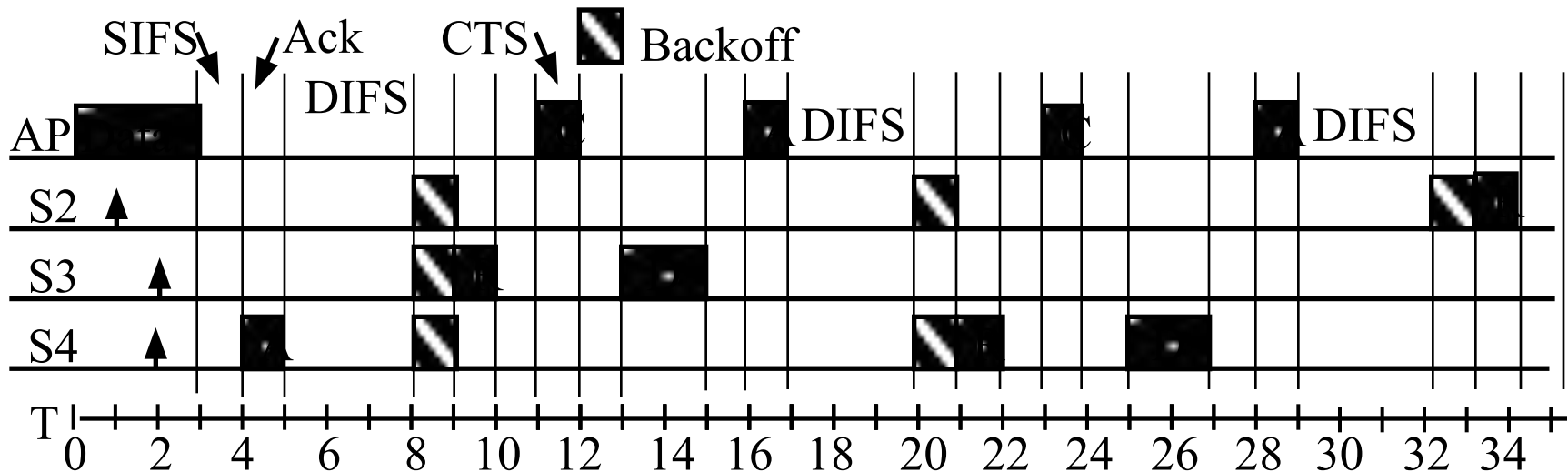
DCF Example

- ❑ Example: Slot Time = 1, CW_{min} = 5, DIFS=3, PIFS=2, SIFS=1
- ❑ T=1 Station 2 wants to transmit but the media is busy
- ❑ T=2 Stations 3 and 4 want to transmit but the media is busy
- ❑ T=3 Station 1 finishes transmission.
- ❑ T=4 Station 1 receives ack for its transmission (SIFS=1)
Stations 2, 3, 4 set their NAV to 1.
- ❑ T=5 Medium becomes free
- ❑ T=8 DIFS expires. Stations 2, 3, 4 draw backoff count between 0 and 5.
The counts are 3, 1, 2

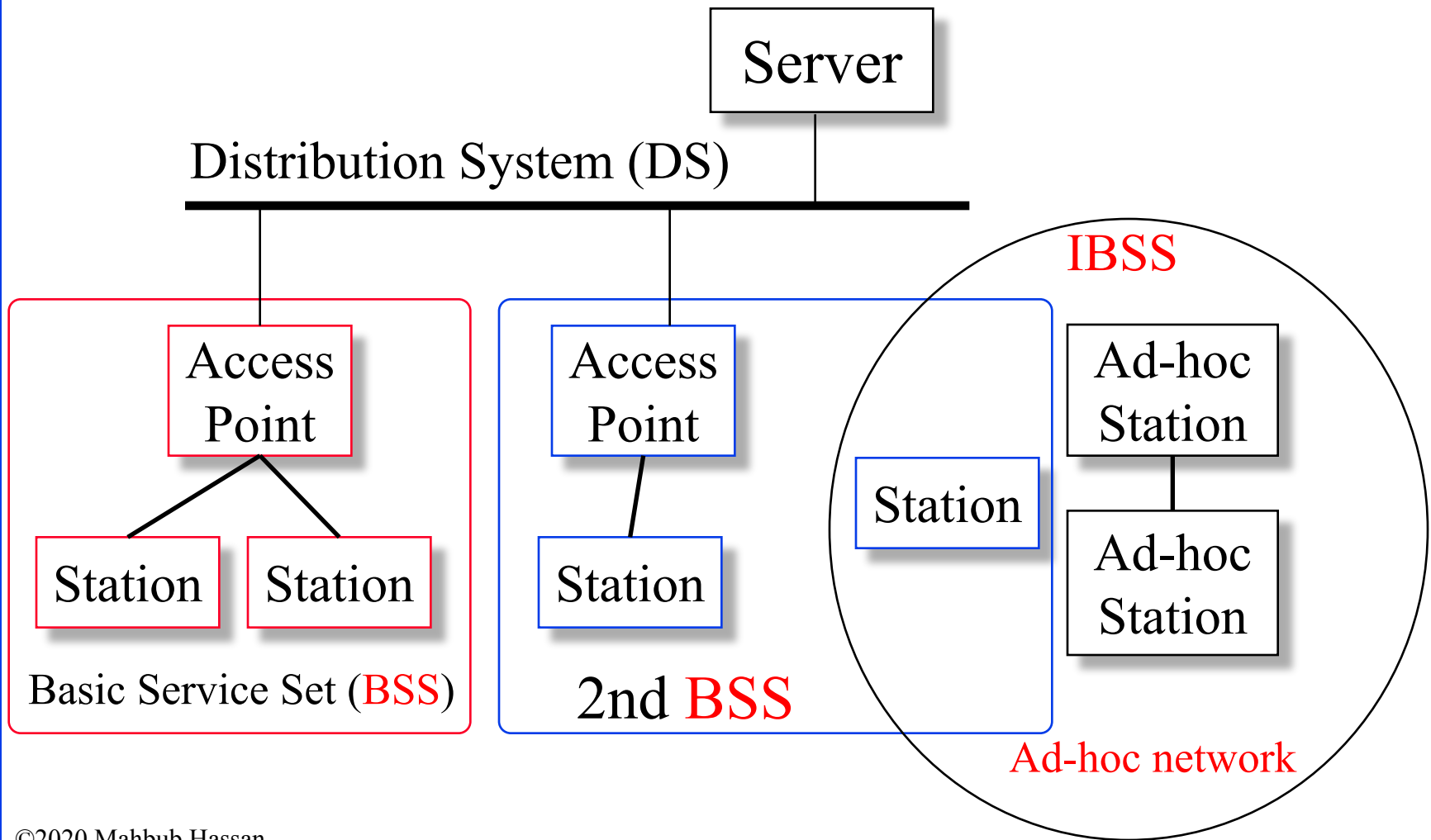


DCF Example (Cont)

- T=9 Station 3 starts transmitting. Announces a duration of 8 (RTS + SIFS + CTS + SIFS + DATA + SIFS + ACK). Station 2 and 4 pause backoff counter at 2 and 1 resp. and wait till T=17
- T=15 Station 3 finishes data transmission
- T=16 Station 3 receives Ack.
- T=17 Medium becomes free
- T=20 DIFS expires. Station 2 and 4 notice that there was no transmission for DIFS. Stations 2 and 4 start their backoff counter from 2 and 1, respectively.
- T=21 Station 4 starts transmitting RTS



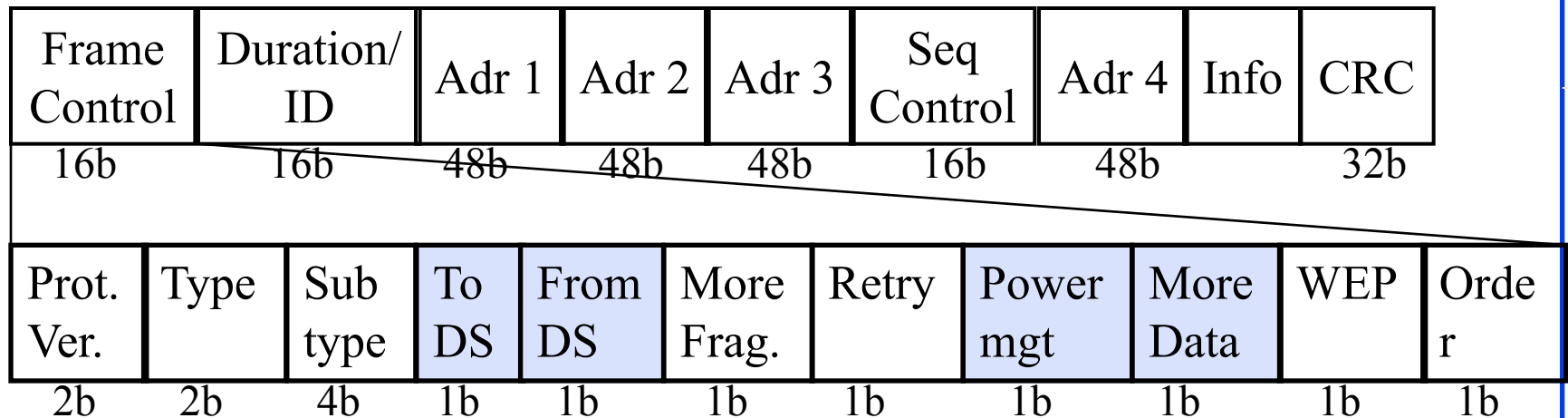
IEEE 802.11 Architecture



IEEE 802.11 Architecture (Cont)

- ❑ **Basic Service Set (BSS)**
= Set of stations associated with **one AP**
- ❑ **Distribution System (DS)** - wired backbone
- ❑ **Independent Basic Service Set (IBSS)**: Set of computers in **ad-hoc mode**. May not be connected to wired backbone.
- ❑ Ad-hoc networks coexist and interoperate with infrastructure-based networks
- ❑ BSSID: 48-bit MAC address of the AP
- ❑ IBSSID: randomly generated address
 - ❑ 2 bits are fixed, 46 bits are generated randomly
- ❑ All-1s BSSID/IBSSID is used for broadcast

Frame Format



- ❑ Type: Control, management, or data
- ❑ Sub-Type: Association, disassociation, re-association, probe, authentication, de-authentication, CTS, RTS, Ack, Power-Save Poll (PS-POLL) ...
- ❑ Retry/retransmission
- ❑ Power mgt: Going to Power Save mode
- ❑ More Data: More buffered data at AP for a station in power save mode
- ❑ WEP: Wireless Equivalent Privacy (Security) info in this frame
- ❑ Order: Strict ordering

MAC Frame Fields

❑ Duration/Connection ID:

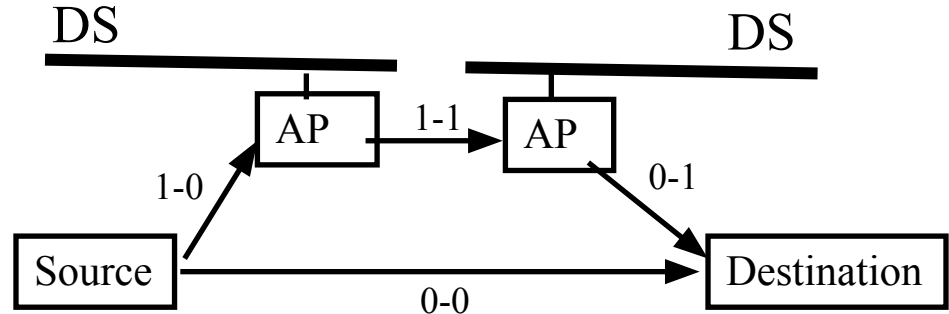
- ❑ If used as duration field, indicates time (in μs) channel will be allocated for successful transmission of MAC frame. Includes time until the end of Ack
- ❑ In some control frames, contains association or connection identifier

❑ Sequence Control:

- ❑ 4-bit fragment number subfield
 - ❑ For fragmentation and reassembly
- ❑ 12-bit sequence number
- ❑ Number frames between given transmitter and receiver

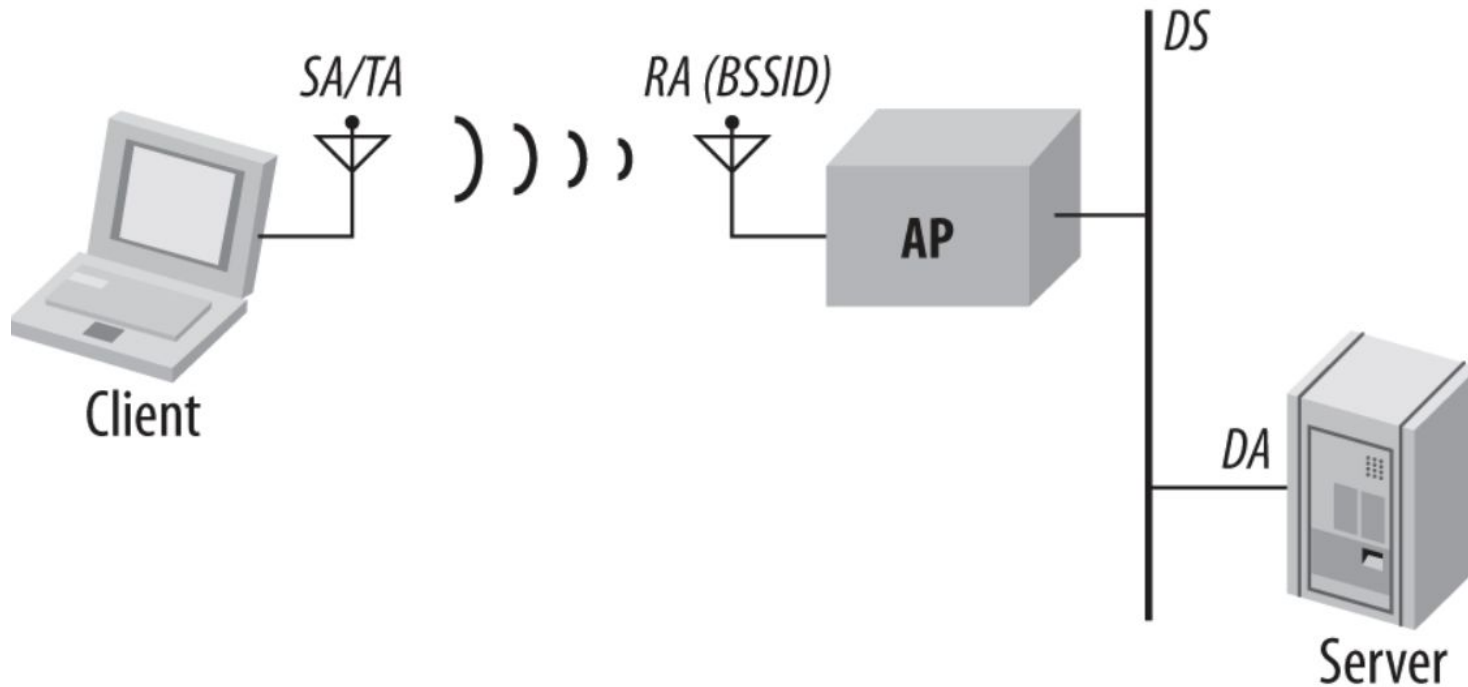
802.11 Frame Address Fields – data frames

- ❑ **Source/Destination:** ultimate network devices that prepare and decode the frame for network layer
- ❑ **Transmitter(Tx)/Receiver(Rx):** Could be the source/destination, or intermediate radio devices, e.g., access point (AP)
- ❑ 4 address fields; defined by 2 DS bits



Purpose	ToDS	FromDS	ADR1 (Rx)	ADR2 (Tx)	ADR3	ADR4
IBSS	0	0	DA	SA	IBSSID	N/A
From AP (from infra.)	0	1	DA	BSSID	SA	N/A
To AP (to infra.)	1	0	BSSID	SA	DA	N/A
AP-to-AP (W'less Brdg)	1	1	RxA	TxA	DA	SA

Example 802.11 Addressing: Wireless Client to Server



Addresses in frames transmitted by the client radio

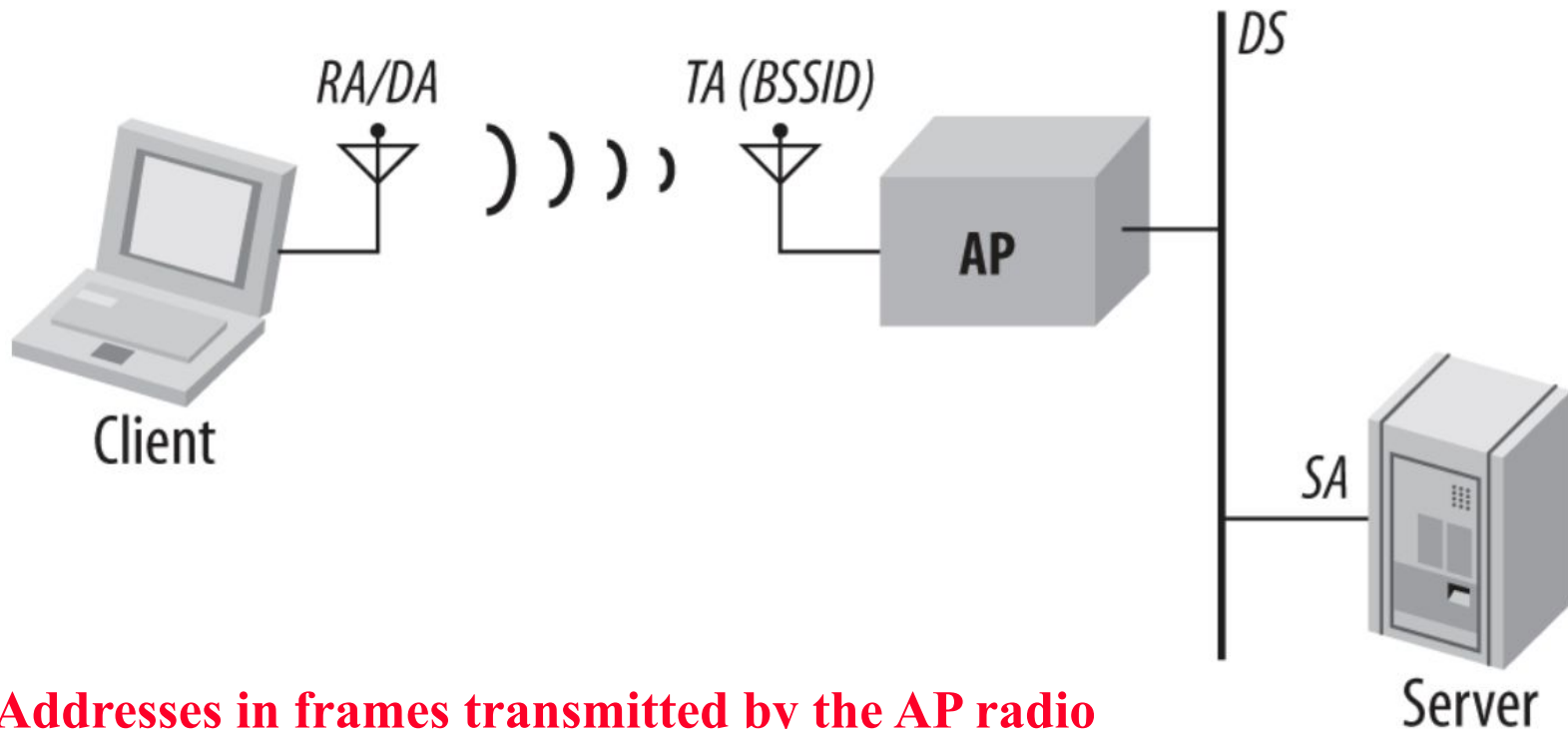
ADR1: AP MAC address (BSSID)

ADR2: Client MAC address (source address)

ADR3: Server MAC address (destination address)

ADR4: Not applicable

Example 802.11 Addressing: Server to Wireless Client



Addresses in frames transmitted by the AP radio

ADR1: Client MAC address (destination address)

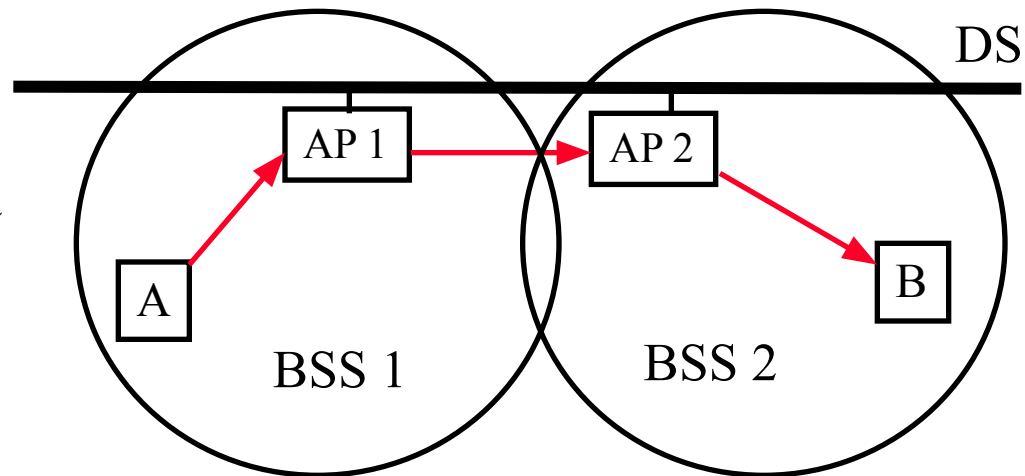
ADR2: AP MAC address (BSSID)

ADR3: Server MAC address (source address)

ADR4: Not applicable

Example 802.11 addressing

- Consider the example WLAN in the figure where two BSSs are connected via a distribution system. What is the content of the **Address 3 field** when Station A wants to send a packet to Station B via AP 1?

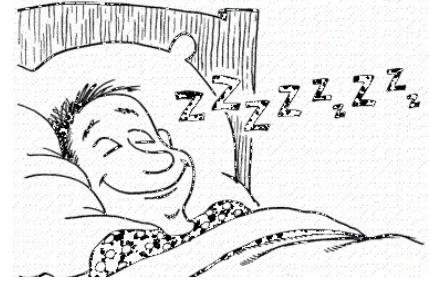


- In this case (To DS=1, From DS=0), Address 3 field should contain the address of the destination station. Therefore, it should be the address of B.

Power Saving

- ❑ Extending the battery life of portable devices is one of the main challenges of wireless networks.
- ❑ Mechanisms must be devised to let the device sleep as much as possible and wake up only when it needs to transmit or receive.
- ❑ If there are no packets to be received, a receiver could go to sleep and save battery power.
- ❑ To facilitate this kind of power saving, IEEE 802.11 has a **power management function**.

802.11 Power Management



- ❑ Station tells the base station its mode:
Power saving (PS) or active
 - ❑ Mode changed by **Power Mgmt bit** in the frame control header.
- ❑ All packets destined to stations in PS mode are buffered (at AP)
- ❑ AP broadcasts list of stations with buffered packets in its beacon frames:
Traffic Indication Map (TIM)
- ❑ When a station wakes up, it waits for the beacon; sends a PS-Poll message to AP if its bit is turned on in TIM; AP then sends one frame with buffered data and sets the **More Data bit** in the header if more data in the buffer (station does not go back to sleep after receiving one frame if **More** is set).

Traffic Indication Map (TIM)

- ❑ A bit map inside a beacon
- ❑ 2008 bits; each bit represents an Association ID (one associated client)
- ❑ If packets are buffered in the AP for a given Association ID, its corresponding bit is set to '1', '0' otherwise

Summary

1. 802.11 PHYs: Spread spectrum in earlier versions, but OFDM in new versions
2. 2.4 GHz channels (22 MHz) are mostly overlapped, but 5 GHz channels (20 MHz) are non-overlapped, but some are shared with the radar service
3. High speed applications can be supported by combining multiple adjacent channels into single channel with higher bandwidth
4. 802.11 uses SIFS, PIFS, DIFS for priority
5. WLAN frames have *four* address fields
6. 802.11 supports power saving mode

Homework 2 part 2

- Two 802.11 stations get frames to transmit at time $t=0$. The 3rd station (AP) has just finished transmitting data for a long packet at $t=0$ to Station 1. The transmission parameters are: Slot time=1, SIFS=1, DIFS=3, $CW_{min}=5$, $CW_{max}=7$. Assume that the pseudo-random number generated are 1, 3. The data size for both stations is 3 slots. Draw a transmission diagram. At what time the two packets will get acknowledged assuming no new arrivals.

Acronyms

☐	Ack	Acknowledgement
☐	AP	Access Point
☐	APSD	Automatic Power Save Delivery
☐	BO	Backoff
☐	BSA	Basic Service Area
☐	BSS	Basic Service Set
☐	BSSID	Basic Service Set Identifier
☐	CA	Collision Avoidance
☐	CD	Collision Detection
☐	CDMA	Code Division Multiple Access
☐	CFP	Contention Free Period
☐	CRC	Cyclic Redundancy Check
☐	CSMA	Carrier Sense Multiple Access
☐	CTS	Clear to Send
☐	CW	Congestion Window
☐	CWmax	Maximum Congestion Window

Acronyms (Cont)

- ❑ CWmin Minimum Congestion Window
- ❑ DA Destination Address
- ❑ DCF Distributed Coordination Function
- ❑ DIFS DCF Inter-frame Spacing
- ❑ DS Direct Sequence
- ❑ ESA Extended Service Area
- ❑ ESS Extended Service Set
- ❑ FH Frequency Hopping
- ❑ FIFO First In First Out
- ❑ GHz Giga Hertz
- ❑ IBSS Independent Basic Service Set
- ❑ ID Identifier
- ❑ IEEE Institution of Electrical and Electronics Engineers
- ❑ IFS Inter-frame spacing
- ❑ ISM Instrumentation, Scientific and Medical
- ❑ LAN Local Area Network

Acronyms (Cont)

❑	MAC	Media Access Control
❑	MHz	Mega Hertz
❑	MIMO	Multiple Input Multiple Output
❑	NAV	Network Allocation Vector
❑	OFDM	Orthogonal Frequency Division Multiplexing
❑	PCF	Point Coordination Function
❑	PHY	Physical Layer
❑	PIFS	PCF inter-frame spacing
❑	PS	Power saving
❑	QoS	Quality of Service
❑	RA	Receiver Address
❑	RTS	Ready to Send
❑	SA	Source Address
❑	SIFS	Short Inter-frame Spacing

Acronyms (Cont)

- ❑ SS Subscriber Station
- ❑ TA Transmitter's Address
- ❑ TIM Traffic Indication Map
- ❑ WiFi Wireless Fidelity
- ❑ WLAN Wireless Local Area Network